

Digoxin Overdose



Section I: Scenario Demographics

Scenario Title:	Digoxin Overdose		
Date of Development:	30/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To push junior learners to the limits of their knowledge and understanding by presenting an overdose case with fairly complex management.
CRM Objectives:	Communicate effectively with team members during management of infrequent presentation.
Medical Objectives:	1) Recognize potential for digoxin toxicity in elderly patient on digoxin. 2) Recognize dysrhythmias associated with a digoxin overdose. 3) Demonstrate an approach to digibind dosing for treatment of digoxin overdose.

Case Summary: Brief Summary of Case Progression and Major Events

90 year-old woman is brought to ED by her daughter because of confusion. She recently had a bout of vomiting and diarrhea and hasn't been taking much PO since. Today, she is less responsive, seems confused, and is complaining of being dizzy. The team will be given a copy of the patient's medication list, which will include digoxin. On arrival, the patient will be hypotensive and her rhythm will be bi-directional VT. Ideally, the team should give digibind. If they do not, they will receive blood work back with a high level to trigger administration.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



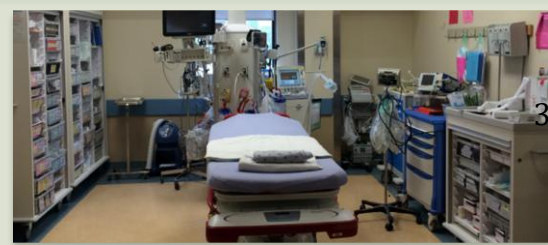
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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
None required.			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None required.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Mildred Funk is a 90 year old woman who is brought to the ED by her daughter because of confusion. She had some vomiting and diarrhea recently and hasn't been eating or drinking much since. Today, she seems confused and keeps complaining that she's dizzy to her daughter.

B. Patient Profile and History

Patient Name: Mildred Funk	Age: 90	Weight: 60kg
Gender: ☐ M ☒ F	Code Status: Full.	

Chief Complaint: "Dizzy"

History of Presenting Illness: Vomiting and diarrhea for the last few days. Now not taking much PO. Confused today, and complaining of being dizzy. **No recent trauma. No travel. No sick contacts.**

Past Medical History:	A fib	Medications:	Coumadin 4mg daily
	CAD (stent x2 in 2009)		Metoprolol 25mg BID
	HTN		Ramipril 5mg daily
	Dyslipidemia		Rosuvastatin 10mg daily
	Hypothyroidism		Levothyroxine 125mcg daily
			Digoxin 0.0625 mg daily

Allergies: None.

Social History: Lives with her daughter. Non-smoker. No EtOH.

Review of Systems:	CNS:	Confused. Complains of being dizzy. No headache.		
	HEENT:	Nil.		
	CVS:	No CP. No palps.		
	RESP:	No SOB.		
	GI:	Nausea. Emesis this morning. Diarrhea last few days.		
	GU:	No LUTS.		
	MSK:	No sore joints.	INT:	No rashes.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 160/min	BP: 90/55	RR: 12/min
O ₂ SAT: 95%	Rhythm: Bigeminy	T: 36.5°C
Glucose: 6.4 mmol/L	GCS: 13 (E4 V4 M5)	
General Status: Confused and complaining of being dizzy.		
CNS:	GCS 13 (confused). No FND.	
HEENT:	No signs trauma.	
CVS:	Tachycardic. No murmur.	
RESP:	GAEB. No adventitious.	
ABDO:	Soft, NT.	
GU:	Nil.	
MSK:	No signs trauma.	SKIN: Nil.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Bigeminy on monitor (ECG: bidirectional VT) HR: 160 BP: 90/55 RR: 12/min O ₂ SAT: 95% T: 36.5°C	Awake, but confused. Answers questions inappropriately with "dizzy."	<u>Learner Actions</u> ☐ IV & monitors ☐ Attach defib pads ☐ ECG ☐ NS 1L bolus ☐ Cardiac/abdo/lactate/VBG and dig level ☐ Hx and Px ☐ Cap sugar: 6.4 ☐ Call for digibind ☐ Call poison control	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - NS bolus → BP 85/45 - No dig level sent → RN to prompt re: med list - Rhythm not recognized: RN to prompt <u>Triggers</u> <i>For progression to next state</i> - Cardioversion → 2. Pulseless - Digibind ordered → 3. Blood Work Back - 5 minutes → 2. Pulseless
2. Pulseless Rhythm → VT HR → 180 BP → ?/?	Pulseless and unresponsive.	<u>Learner Actions</u> ☐ Apply defib pads ☐ Shock VT ☐ Give digibind (10-20 vials) ☐ Call poison control ☐ Epinephrine q3min ☐ High quality CPR ☐ ± Amiodarone (if don't recognize dig toxicity) ☐ ± Intubation	<u>Modifiers</u> - Shock → No change to rhythm <u>Triggers</u> - Digibind given → 3. Blood Work Back - 7 min → 3. Blood Work Back
3. Blood Work Back **Same vitals as previous state	State begins by saying "blood work is back" and giving results showing dig level 8	<u>Learner Actions</u> ☐ Calculate digibind dose # vials = [serum dig level (ng/mL) x weight in kg]/100 = [8x60]/100 = 4.8 = approximately 5 vials **Because chronic toxicity, can assume dig level is steady state level ☐ Call poison control	<u>Modifiers</u> <u>Triggers</u> - Digibind given → 4. Stabilization - 12 min → 4. Stabilization
4. Stabilization Rhythm → a fib HR → 120 BP → 100/60		<u>Learner Actions</u> ☐ Call medicine/ICU ☐ Repeat ECG ☐ Reassess patient	END CASE → Consultants Arrive

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 129	K: 3.2	Cl: 90	HCO ₃ : 12	BUN: 15	Cr: 132	Glu: 6.3
Ca: 2.1		Mg: 0.91		PO ₄ : 1.4		Albumin: 30
VBG	pH: 7.30	PCO ₂ : 24	PO ₂ : 45	HCO ₃ : 12	Lactate: 2.6	
WBC: 12		Hg: 144		Hct:		Plt: 350
Digoxin: 8 ng/mL		Troponin I: 25		INR 2.5		aPTT: 35
TB: 8		AST: 30		ALP: 40		GGT: 24
Lipase: 20				APAP: <20		ASA: <0.36

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Images (ECGs, CXRs, etc.)

ECG – Bidirectional VT



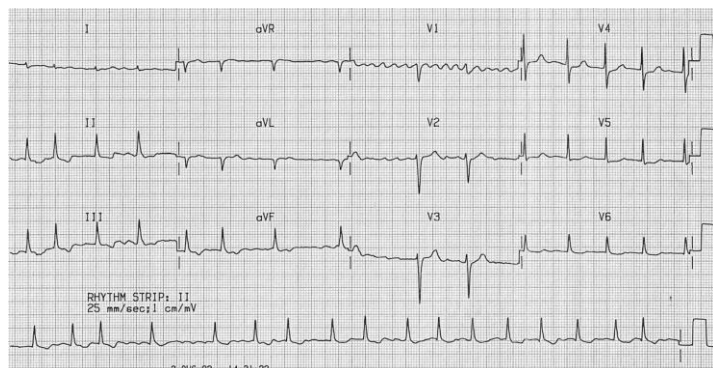
<http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/04/Bidirectional-VT.jpg>

CXR – normal



<http://radiopaedia.org/articles/normal-position-of-diaphragms-on-chest-radiography>

ECG – moderately rapid a fib



<http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/08/af1.jpg>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) When did you first consider digoxin toxicity as a possible cause of this presentation? 2) What clues were there in this case that pointed toward digoxin toxicity? 3) Did your team feel comfortable calculating a digibind dose? How did your team dynamics feel with this uncommon medication to administer? 4) What are the indications for digibind? Did this patient meet them? 5) What are other important differential diagnoses to consider in this elderly patient with altered LOC?	
Key Moments	
Recognition of possible digoxin toxicity	
Determination of digibind dose	